

Field manipulation of electromagnetically induced transparency analogue in terahertz metamaterials for enhancing liquid sensing

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ABSTRACT

Terahertz (THz) metamaterial sensing, an emerging technology in biomedical sciences, has made considerable progress with unique characteristics, such as label-free, non-invasive, non-destructive, and efficient. However, excellent sensing performance involves the need for extremely strong light-matter interaction that is difficult to implement for the intrinsic divergent electric field excited on the surface of metamaterials. Here, an innovative method of manipulating the electric field to realize maximum light-matter interaction for enhancing liquid sensing at THz frequencies is reported. The electric field of electromagnetically induced transparency (EIT) resonance is tightly confined inside the structure around the quasi-dark mode by optimizing the fully-enclosed metamaterial structure design. More importantly, the electric field is enhanced and the part that permeates into the substrate is exploited, together with the formation of inartificial fluid loading units by the effective regional substrate etching, exhibiting the enhanced interaction between the polar liquids and the THz waves. The measurements reveal a significantly improve in sensitivity of 0.312 THz/RIU, which is 9.8 times that without etching by contrast. The strong light-matter interaction implements in THz metamaterial via the electric field manipulation holds bright promises for realizing highly sensitive THz sensing, forging a route for real-time supervising trace biomolecules.

1. Introduction

The terahertz (THz) radiation with the richness of photonics and electronics presenting between 0.1 and 10 THz has attracted considerable attention [1–3], which favors their application in communication [4,5], security screening [6], non-destructive [7], imaging [8] and biomedical sciences [9,10]. In particular, the low photon energy of THz pulse, together with characteristic molecular fingerprint information [11], driving THz technology is expected to be an emerging wavelength regime for investigating molecular information [12]. However, THz spectroscopy in molecular detection at low concentrations remains hindered by minuscule absorption cross-sections caused by the scale mismatch between THz wavelength and molecule. Recently, the metamaterial with peculiar electromagnetic response has emerged as one of the most powerful platforms for improving sensitivity with the help of highly localized electromagnetic field [13–19]. The unique characteristics of metamaterials are derived from the ability to compress electromagnetic fields [20–22], which is conducive to strengthening the

light-matter interaction between the THz wave and the analytes [23], and stimulates researchers to exploit THz metamaterials for biochemical sensing [24–26]. However, their remarkable advantages aside, most of the existing THz metamaterial sensors remarkably suffer from a poor limit of low sensitivity, resulting in powerless detection of the trace analyte or the faint changes of medium environment.

Newly, successful examples of periodic subwavelength metallic structures had been introduced for high sensitivity THz metamaterial design, such as two symmetric proximal resonators [27], cross shaped absorber [28], asymmetric split ring [29,30], grooves array [31,32], nanoslot array [33], asymmetric tetramer clusters [34]. Generally, most of these structures have assisted in improving the performance of the sensor to a certain extent. In addition, the concept of electromagnetically induced transparency (EIT) which is a quantum interference effect that forms a narrow transparent window creates another opportunity to address these drawbacks [35–37]. In such kind of resonance effect, the destructive interference between two excitation pathways makes metamaterial possess the ability to detect weak changes of external dielec-

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tronic environment [38,39]. To capitalize on these sensitive properties, practical and accurate screening of the oral cancer cells HSC3 [40] and glioma cells [41] had been reported based on this technology. However, most metamaterials rely on innate electric field excited by the surface [25,27,29,39–43], which is not most preferably and restricts the sufficient use of the electric field. The high sensitivity of metamaterials is derived from the highly localized electromagnetic fields in the subwavelength region and the optimal overlap of the analyte and the excited electromagnetic fields [44,45]. Although, some of the researchers had demonstrated a significant improvement of sensing performance by decreasing the influence of the substrate [44–49]. However, the substrate etching in part will result in inadequate use of the electric field, and in whole will augment futile space where no electric field excitation, which impeding its efficient application in liquid environment with high absorption in THz spectral domain.

In this work, we experimentally demonstrated an EIT metamaterial containing a closed square ring and U-shaped split ring resonators for polar liquids sensing. Benefiting from the bright mode being suppressed in EIT analogue, the strongly concentrated electric field distributes around the quasi-dark mode. By reasonable local substrate etching around the quasi-dark mode, the part of the electric field that permeates into the substrate which is often disregarded can be utilized. More importantly, the electric field is enhanced by the decrease of effective dielectric constant, which maximizes the interaction between the polar liquid analytes and the electric field, while diminishes the absorption loss possible. Moreover, the metamaterial with fully-enclosed structure possesses fluid loading units formed by local substrate etching, integration with the sample cell to support enhancing liquid sensing. Both experimental and numerical results show dramatic increases in sensitivity contrasting against the original structure. The proposed THz enhanced sensing method by manipulating the electric field has universality and portability to other structures, and enables a novel platform to realize highly sensitive sensors for polar liquids sensing applications.

2. Results and discussion

The schematic diagram shown in Fig. 1(a) highlights the core idea of this work, where the electric field concentration region of the metamaterial array is etched to maximize the light-matter interaction. The metamaterial consists of a U-shaped split ring resonator (U-SRR) surrounded by a concentric metallic closed square ring resonator (CRR). The designed EIT metamaterial structure with and without substrate etching was fabricated on a 2-inch silicon wafer with 300 μm thickness. The resistivity of silicon is 10k $\Omega\cdot\text{cm}$, due to high-resistivity silicon has lower loss and dispersion than low-resistance silicon in the terahertz band of interest for this study [17]. Firstly, after plasma surface treatment, 200 nm thick aluminum was sputtered onto the silicon wafer. Secondly, photoresist AZ5214 was coated on the silicon wafer by spin coating process followed by a soft-baking at 95°C for 90 s. Then, the photoresist was exposed to UV light for patterning using a proper mask with an exposure time of 6.5 s, and developed with TMAH (2.38%) developer for 45 s. Next, excess aluminum was removed by ion beam etching (IBE) to reveal the metamaterial array without etching. And after the first time of photoresist cleaning by acetone and isopropanol, photoresist AZ6130 was spin-coated on the processed silicon wafer and adopted the same lithography steps (slightly different in time) as before to expose the etched area on the other half of the silicon wafer. The next step was silicon substrate etching by deep reactive ion etching (DRIE) using Bosch process. Finally, after the second time photoresist cleaning, the wafer was cut into 1.7 cm $\times$ 1.7 cm samples of with and without substrate etching by a dicing saw. The scanning electron microscope (SEM) images of the fabricated metamaterial arrays without and with 15 μm depth substrate etching is depicted in Figs. 1(b) and 1(c), which indicate the excellent structural quality of the fabricated samples. With grey substrate severely what to contrast is purely black etched area more clearly demonstrated in Figs. 2(a) and 2(b). The unit cell shown in Figs. 1(d) and

1(e) show the crucial dimension parameter. The unit cell has a square periodicity period of 60 μm along both axial directions. The U-SRR has the length of side, $s = 32 \mu\text{m}$, and the CRR is $l = 50 \mu\text{m}$. Both have a width of the line, $w = 3 \mu\text{m}$. In addition, to ensure the processing quality as far as possible, avoid excessive distortion of the verticality, and meet the optimal performance of the sensor, the appropriate etching depth was set to 15 μm .

In EIT analogue, the bright mode almost be suppressed and only the quasi-dark mode is enhanced due to the strong interference between the resonators. In order to manipulate the electric field to be compressed in a more concentrated space, the CRR is selected as the bright mode to limit the electric field in the center like a boundary. The excitation of EIT analogue in designed metamaterial structure is achieved by coupling between the CRR and U-SRR (see the Supporting Information for formation of the EIT analogue), which have resonances fall at the same frequency and possess highly contrasting Q -factor (Q -factor is defined as $Q = f / \text{FWHM}$). The transmittance spectrum of the EIT metamaterial was obtained by [17]:

$$E_{\text{sam}}(\omega) / E_{\text{ref}}(\omega) = A(e^{i\phi}) \quad (1)$$

$$T(\omega) = |A(\omega)|^2 \quad (2)$$

where $E_{\text{sam}}(\omega)$ and $E_{\text{ref}}(\omega)$ are fast Fourier transform (FFT) of the time-domain THz pulses of the sample and reference respectively, which are obtained by the commercial terahertz time-domain spectroscopy (THz-TDS) system (TeraPulse 4000) in a nitrogen-purged chamber. The THz time-domain pulse generated by the system has a pulse width of about 4 ps and an available bandwidth from 0.2 to 2.5 THz. Femtosecond laser as the excitation source with a pulse width of 100 fs, a repetition rate of 100 MHz and an output power of over 65 mW at a center wavelength of 780 nm, irradiating on GaAs photoconductive antenna to realize the generation and detection of THz radiation. The experimentally measured transmission spectra from the EIT metamaterial both with and without the 15 μm deep etching are shown in Fig. 2(c). For comparison, the simulated transmission spectra before etching and at different etching depths are shown in Fig. 2(d). The numerical simulations were carried out using CST Microwave Studio frequency domain software, where the unit-cell boundary condition was imposed in the x and y directions and open boundary in the z-direction. At least 10 mesh steps per-wavelength were employed to assure the accuracy of the simulated results. In simulation, the aluminum was defined as lossy metal with a conductivity of $3.56 \times 10^7 \text{ S m}^{-1}$, and silicon was modeled as a lossless dielectric with $\epsilon_{\text{si}} = 11.9$. The experimental dates before and after 15 μm etching show reasonably well agreement with the simulation results and reproduce all the spectral features in the spectral region of interest. The subtle differences in resonant frequency and amplitude may attributed to process tolerances. Clearly, the measured EIT resonance blue-shifts from 0.662 to 1.132 THz and increases in amplitude meanwhile, where the internal regions of the metamaterial structure are etched. The blue-shift is derived from the decrease in the effective dielectric constant can be further explained by coupling between two sub-resonators (later section).

Furthermore, we apply the coupled harmonic oscillator model to phenomenologically explain the coupling between the CRR and U-SRR elements as the etching depth (t) changes. And the resonators can be regarded as both particles (bright and quasi-dark modes), and the coupling equations are as follows [50–52]:

$$\begin{aligned} \ddot{x}_b(t) + \gamma_b \dot{x}_b(t) + \omega_b^2 x_b(t) + \kappa^2 x_d(t) &= \frac{Q}{M} E \\ \ddot{x}_d(t) + \gamma_d \dot{x}_d(t) + \omega_d^2 x_d(t) + \kappa^2 x_b(t) &= \frac{q}{m} E \end{aligned} \quad (3)$$

Here, bright and quasi-dark modes are described as particles b and d. The particles with resonance angular frequency (ω_b , ω_d) and loss factors (γ_b , γ_d) are described by the excitation ($x_b(t)$, $x_d(t)$), and κ is the coupling strength of the two particles. We substitute the relative coupling of

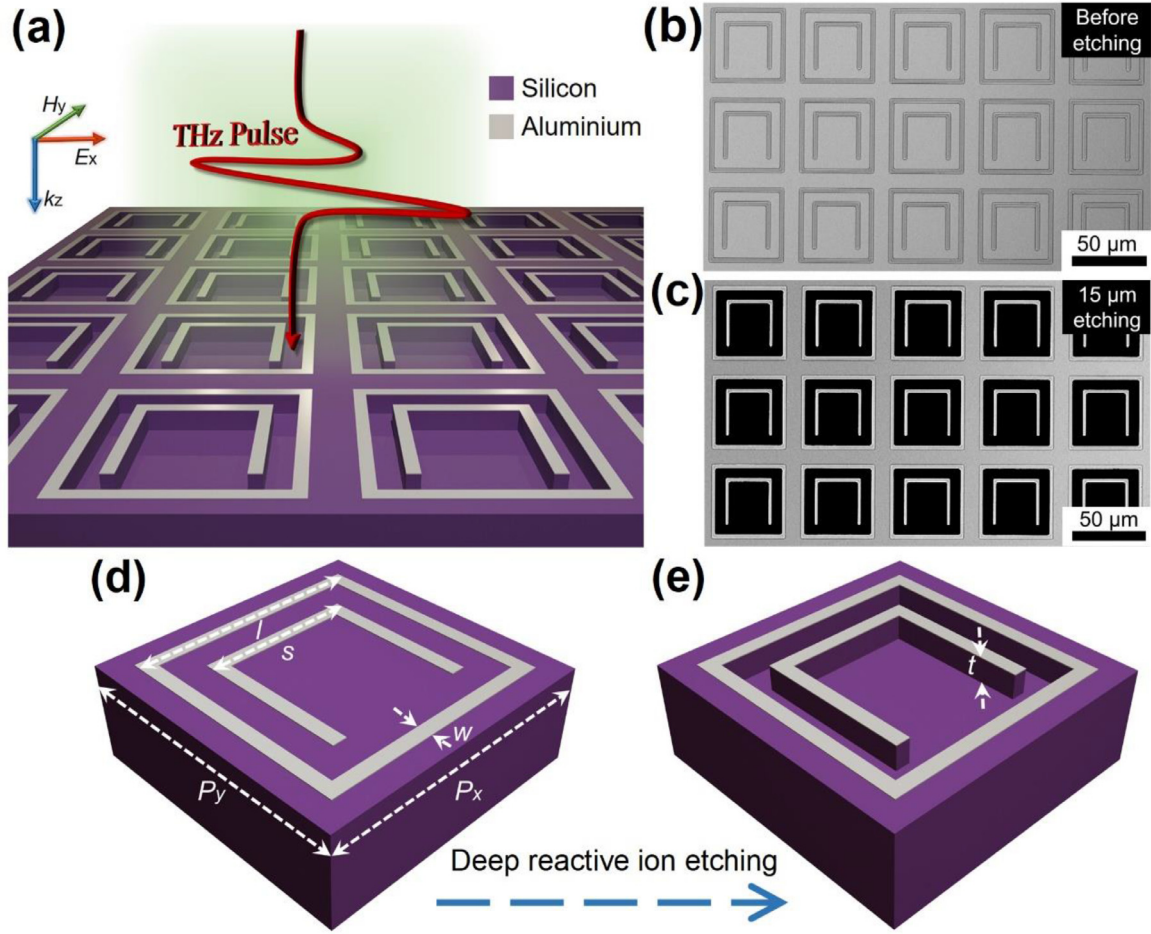


Fig. 1. Illustration of the proposed THz metamaterial. (a) Schematic diagram of the EIT metamaterial consists of a square array of aluminum CRR and U-SRR with 15 μm depth local silicon substrate etching. The etching area is located around the U-SRR inside the structure where the electric field is excited at EIT resonance. Scanning electron microscope image of the fabricated EIT metamaterial arrays (b) without and (c) with substrate etching. The unit cell dimensions are depicted in (d) and (e), where $p_x = p_y = 60 \mu\text{m}$, $l = 50 \mu\text{m}$, $s = 32 \mu\text{m}$, $w = 3 \mu\text{m}$ and $t = 15 \mu\text{m}$, respectively. The thickness of aluminum is 200 nm.

incident radiation with bright and quasi-dark modes using dimensionless constants $A = Q/q$ and $B = M/m$, where (Q, q) and (M, m) are their effective charge and effective mass, respectively. The effective susceptibility χ_{eff} of the system, which correlated the polarization (P) of the particle to the strength of incident electric field (E), can be obtained by solving the above coupled Eq. (3) for x_b and x_d :

$$\chi_{\text{eff}} = \frac{P}{\epsilon_0 E} = \frac{Qx_b + qx_d}{\epsilon_0 E} = \frac{K}{A^2 B} \left(\frac{A(B+1)\kappa^2 + A^2(\omega^2 - \omega_d^2) + B(\omega^2 - \omega_b^2)}{\kappa^4 - (\omega^2 - \omega_b^2 + i\omega\gamma_b)(\omega^2 - \omega_d^2 + i\omega\gamma_d)} + i\omega \frac{A^2\gamma_d + B\gamma_b}{\kappa^4 - (\omega^2 - \omega_b^2 + i\omega\gamma_b)(\omega^2 - \omega_d^2 + i\omega\gamma_d)} \right) \quad (4)$$

The transmission can be obtained by $T = 1 - \text{Im}(\chi_{\text{eff}})$ (given by the Kramer-Kronig relations), where χ_{eff} is the absorption (loss) in the medium. We plot the analytically modeled transmission spectra, as shown in Fig. 3(a)-3(d), which show good agreements with simulated and experimental results for the EIT system with varying t , all exhibiting a blue-shift of the resonance. The slight deviations in linewidth and amplitude may originate from weak manufacturing imperfections and the impact related to the periodicity of metamaterial [53,54].

The fitting parameters with different t are plotted in Fig. 3(e). It is noticed that ω_d shows a drastic blue-shift from 0.635 to 1.11 THz as t increases, the variation is evident at first and gradually decreases until ω_d towards stability, which is related to the penetration depth of the electric

field. A similar trend is observed for ω_b , while its sensitivity to t is much lower than that of ω_d attributed to the choice of etching area. For the LC resonance excited by the U-SRR, its resonance frequency (ω_d) strongly depends on the dielectric permittivity of the hotspots where the electric field is excited [47]. In addition, CRR possesses a dipole resonance with a resonance frequency at $\omega_b = 1/(2d\sqrt{\epsilon_{\text{eff}}})$ [55], where d is the equivalent length of the metamaterial and ϵ_{eff} is the effective dielectric constant. The observed blue-shift in the frequencies of bright and quasi-dark resonators in the coupled system is due to the reduction of the effective permittivity by substrate etching. In addition, γ_b and γ_d are almost identical with the increase of t , whereas the coupling strength κ has a slight increase from 0.286 THz without etching to 0.49 THz at 15 μm of etching depth. The increase in κ leads to more pronounced suppression of radiation losses, which is advantageous for sensing for EIT resonance that only relies on non-radiative damping, since any change that results in line shape will be attributed to changes in the surrounding dielectric environment. From the above, it can be concluded from Fig. 3(e) that the blue-shift of the EIT resonance mainly arises from the change in the resonant frequency of the quasi-dark mode. Furthermore, we show the transmission spectra against t varying from 0 to 20 μm by color map in Fig. 3(f). As observed, a remarkable blue-shift of the EIT peak follows t keeps increasing until etching depth reaches about 18 μm . Notably, during the enlargement of the t , it can be observed that the resonance peak exhibits extreme sensitivity at the beginning which is attributed to the electric field is excited around the metal structure, then gradually diminishing and eventually stabilizing. The resonance peak stops blue-

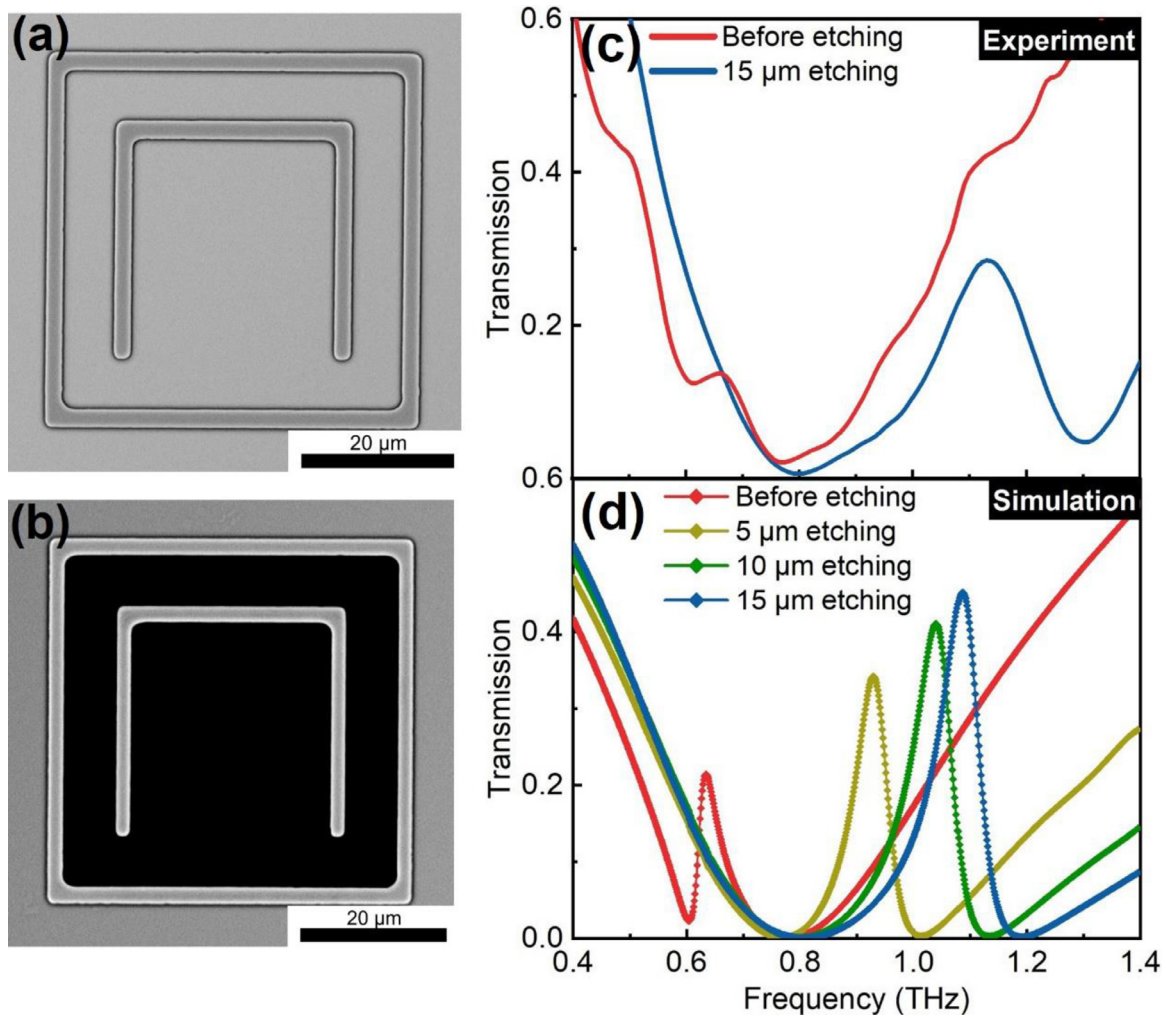


Fig. 2. Experimental demonstration of the proposed EIT metamaterial. The unit cell SEM images of the fabricated EIT metamaterial (a) without and (b) with substrate etching in top view. (c) The experimentally measured THz transmission spectra of the metamaterial before and after 15 μm etching. (d) The numerically simulated THz transmission spectra before etching and at different etching depths.

shift indicates that the weak electric field in the deeper part no longer works, which means the sensing enhancement has reached the optimal. We have slightly discarded the sensor performance and determined the most suitable etching depth to be about 15 μm for ensuring processing perpendicularity without excessive distortion.

Obviously, a metamaterial sensing system with concentrated effect of high-density electric field together with high overlap between confined field enhancement and target substances ensures a strong light-matter interaction is preferred. It is important to note here that the electric field can not only be maximally utilized as a part of it penetrates into the substrate, but can also be enhanced due to the variations in the dielectric environment by substrate etching. In particular, the electric field distributions with and without the 15 μm deep etching at EIT resonance and their different x-z cross-sections shown in Fig. 4(a) exhibit that the electric field is greatly enhanced in the etched region around the metamaterial structure. We further plot the electric field distribution in the hotspot region along the z-axis in Fig. 4(b), the inset shows the concrete position of representative hotspot. It is observed that a high concentration of electric field on the substrate surface and shows an exponential decay as receding from it. The electric field above the substrate does not change significantly by 15 μm etching, whereas the bottom part shows field enhancement markedly attributes to their lower effective permittivity [56]. This is derived from the weaker ability to retain electric charges by etching the silicon substrate around the excited electric

field [57]. The selection of the etched area forms inartificial fluid loading units with tightly confined electric field can provide the maximum overlap between the target substances and the electric field, allowing for highly sensitive optical detection.

Experimentally verified the enhanced sensitivity of the THz metamaterial sensor by field manipulation. The sample cell carefully integrated by clamp, cap and silicon substrate was used to fill liquids [58], the diagram of the sensor in x-z section shows in Fig. 5(f). The silicon substrate is clamped together by two poly-methylpentene (TPX) clamps, above which is lined with EIT arrays. The upper part of the clamps has projecting pillars to support the sample cell sealed by a front bulgy cap constructed from TPX, which is almost transparent to THz pulse. The thickness of two clamps is 6 mm and cap is 5 mm, and the front bulgy of cap is 3 mm. The thickness of the liquid solution is approximately 20 μm , which can be controlled by the clamps. The liquid solution was first dripped onto the surface of the metamaterial chip gripped by the clamps with a pipette, and then the excess liquid was squeezed out by the front bulgy cap to form a sealed sample cell for liquid sensing. A series of fluids like alcohol ($n = 1.45$), glycerin ($n = 1.82$) and water ($n = 2.1$) were selected as sensing analytes, which have a good spread of refractive index values. Figs. 5(a) and 5(b) show the transmission spectra of the EIT metamaterial with and without 15 μm etching for various liquid analytes. An empty sample cell window was served as the reference. It is observed that EIT resonance has a weak red-shift with increasing n from

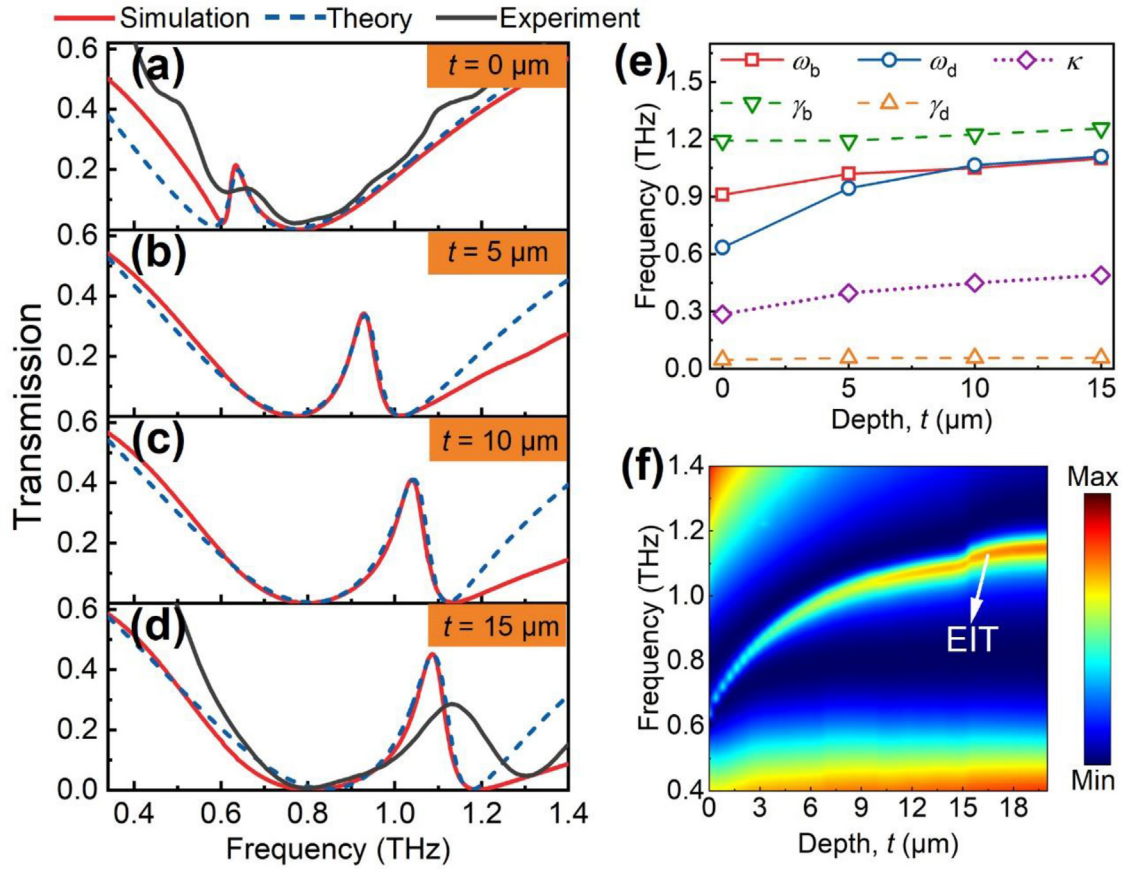


Fig. 3. The blue-shift behaviour of EIT resonance in substrate etching. (a)-(d) Simulated, experiment and corresponding analytical fitted transmission spectra with different etching depth. (e) The fitting parameters ω_b , ω_d , κ , γ_b and γ_d as a function of etching depth. (f) Color map showing THz transmission spectra against frequency and etching depth.

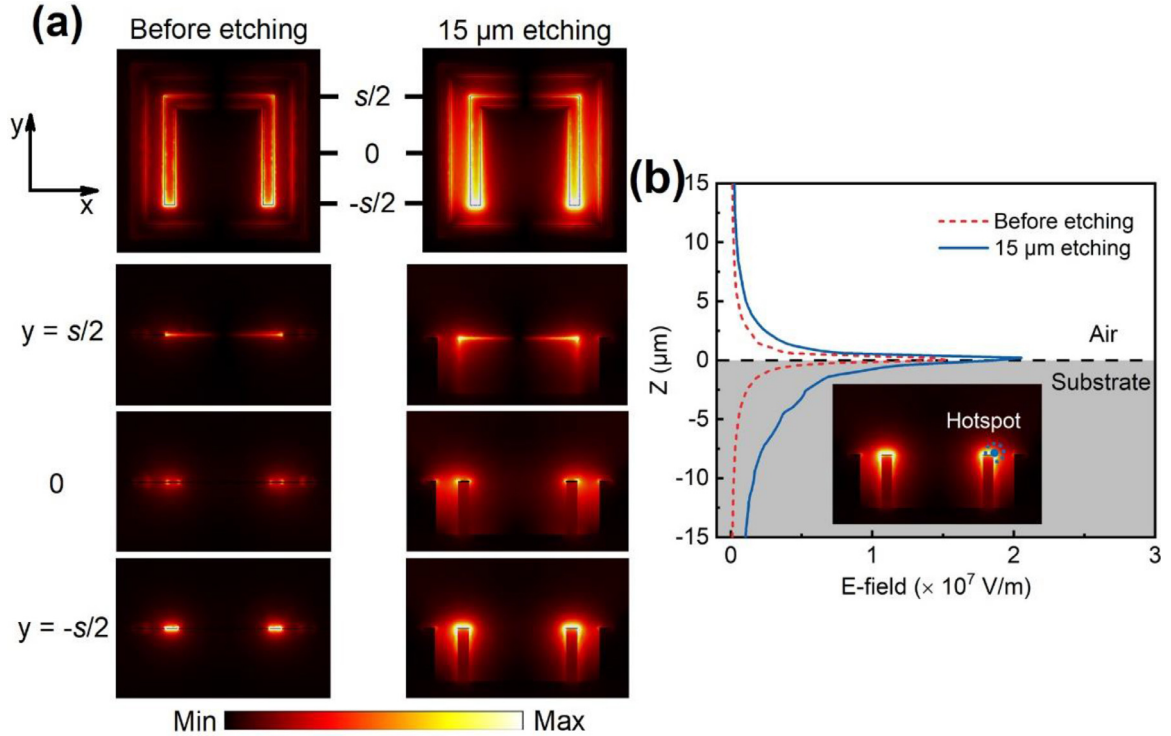


Fig. 4. The electric field enhancement over the design trench. (a) a series of the field distributions at different cross-sections without and with 15 μm substrate etching. (b) The electric field intensity distribution along the z axis at the hotspot region.

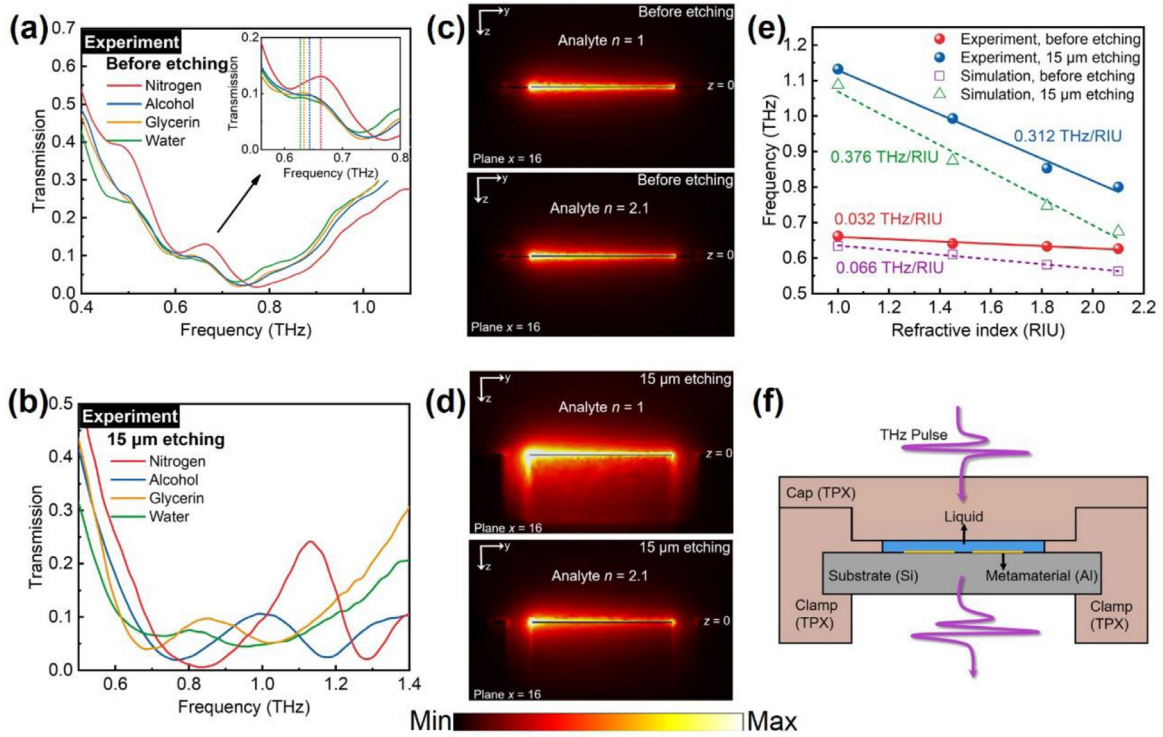


Fig. 5. Experimental results for liquids sensing. Measured transmission spectra of the EIT metamaterial (a) without and (b) with 15 μm substrate etching under varied liquid environments. The electric field distribution in the z -direction without and with analyte for metamaterial (c) before and (d) after 15 μm etching at $x = 16 \mu\text{m}$ cut-plane. (e) The experiments and simulations resonance frequency against the refractive index of liquids and their linear fitting curves. (f) Schematic diagram of the ideal THz metamaterial sensor of the sample cell assembly.

0.662 THz for nitrogen to 0.626 THz for water before etching, and the inset shows the enlarge version of resonant peak. Although analytes with a wide distribution of refractive index values can still be identified by frequency shift, whereas this is not most beneficial for monitoring weak changes in the dielectric environment. After 15 μm substrate etching, a drastic red-shift is observed at EIT resonance from 1.132 THz for nitrogen to 0.8 THz for water, which means a strong interaction between the THz wave and the analyte indicating a sensitive device. We plotted the electric field distribution before and after 15 μm etching when the sensor is empty and filled to account for the increased frequency shift in Figs. 5(c) and 5(d). All electric field plots are cross-sections at $x = 16 \mu\text{m}$, which is the edge of the etch. We observed a distinct field enhancement after 15 μm etching in Fig. 5(d), together with an enhanced degree of overlap between the field and the analyte, which means that a larger proportion of the field has access to changes in the refractive index of the surrounding environment. When analyte with refractive index of 2.1 is filled into the sensor, it can be seen that the field is squeezed dramatically in the z -direction for the etched metamaterial. In contrast, the field which penetrates into the substrate does not show any change in the metamaterial without etching, only the field at the surface shows a weak confine in presence of the analyte [28]. Therefore, the high degree of interaction between the enhanced field and the analyte present greater frequency shifts when the etched metamaterial is filled with different liquid analytes.

A large frequency shift per unit of refractive index means the sensor has superior sensing performance. To better compare the sensitivity of EIT metamaterial with and without 15 μm substrate etching, frequency shifts versus the refractive index of various liquids for experiments are plotted in Fig. 5(e), which shows a linear relationship. The straight lines (red for without etching and blue for 15 μm substrate etching) are linear fits given by $f = -0.032n + 0.692$ and $f = -0.312n + 1.441$ with high R-Square of 0.964 and 0.99. Hence, the absolute slope of the linear fitting curves represents the sensitivity S given by 0.032 THz/RIU and 0.312 THz/RIU, respectively. In contrast, the etched EIT metamaterial

is 9.8 times compared to that without etching. Compared with the performance of reported different sensing approaches (Table S1, Supporting Information), our work shows superior performance in sensitivity, although it does not predominate before etching. Moreover, a similar phenomenon occurs in simulation with the sensitivity of 0.066 THz/RIU and 0.376 THz/RIU get from linear fitting curves for without and with substrate etching. The experimental and simulation results confirm the sensing improvement caused by the electric field enhancement analyses above.

3. Conclusions

In summary, we demonstrated an EIT metamaterial based THz sensing system with high sensitivity via effective electric field manipulation by optimized structural design and local substrate etching. The formation of EIT resonance results from the destructive interference between the near fields of dipole and LC resonances that is interpreted by the surface electric field and current distributions. The metamaterial with fully-enclosed structure design endows the capacity to manipulate electric field where the bright mode be suppressed in EIT analogue. An artificial fluid loading unit with tightly confined electric field is formed by etching the local substrate around the quasi-dark mode. Beyond that, the blue-shift of EIT resonance is observed as the substrate is etched that results from the variation in the resonant frequency of the quasi-dark mode in the near-field coupling of two resonators through the coupled harmonic oscillator model. The electric field is enhanced by decrease in effective permittivity together with the part that permeates into the substrate be utilized can strengthen light-matter interaction in realizing highly sensitive sensing. Benefitting from the overlap of tightly confined local electric field and polar liquids in the fluid loading unit and around the structure, the measurements recorded a significantly high sensitivity of 0.312 THz/RIU at 15 μm etching depth, which is 9.8 times as much as the metamaterial without etching. The reasonable manipulation of electric field is highly suitable for the realization of strong light-matter

interaction to develop high precision platform for identification of low density biomolecules.

Author statement

Tingling Lin - conceptualization, preparation of the manuscript, data curation, and writing of the original draft. **Shuncong Zhong** - conceptualization, funding acquisition, supervision of the work, review, and editing of the manuscript. **Yi Huang and Yujie Zhong** - simulation and analysis. **Zhenghao Zhang and Qiuming Zeng** - proofread and graphic design. **Yingjie Yu and Zhike Peng** - revision and editing of the manuscript. All authors have read and approved the published version of the manuscript.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.optlaseng.2022.107127](https://doi.org/10.1016/j.optlaseng.2022.107127).

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